Polypharmacy Adverse Drug Event (ADE) Cohort Pipeline (ed_non_opioid)

Phase 0: Virginia APCD Longitudinal Lake (2016–2020)

Medical Claims Lake
(All-Payer Encounters, HCG Detail Codes, ICD-10 Positions 1–10)

Pharmacy Claims Lake
(Adjudicated Dispenses, NDC to Generic RxNorm, Days Supply)

Annual Partitions (2016, 2017, 2018, 2019)
Stratified into 8 Age Bands (0–12 to 85–114)

Phase 1: Case Anchor & Opioid Cleanse (phase1_data_preparation.py)

Milliman HCG Clinical Emergency Anchor
Target Codes: P51b (ER), O11 (ER Services), P33 (Outpatient Observation)
Target Timestamp: first_ed_non_opioid_date (t_0)

Absolute Opioid Cleanse Boundary
Strictly drops any patient with ANY opioid ICD-10 code
(F11.* or T40.*) across all 10 diagnosis columns in study year
Guarantees 100% mutual exclusivity with opioid cohort

Phase 2: Sequential ADE Filtering Funnel (phase2_event_processing.py)

Filter 1: ED Utilization Cap (< 7 ED Visits/Year)
- **Excludes chronic crisis utilizers ('flyers')**
- **Removes severe socio-environmental crisis noise**
- **Isolates true acute medication-mediated emergencies**

Filter 2: Proximal Drug Exposure Window (1–21 Days Pre-ED)
- **Patient must have active pharmacy dispensing 1–21 days prior**
- **FDA FAERS Pharmacokinetic Validation (qa_faers_time_to_onset.py):**
21-day window captures ~90.5% of all acute ADE time-to-onset distributions

Filter 3: Zero-Day Discharge-Fill Guard
- **STRICTLY PURGES all prescriptions dispensed on Day 0 (t_0)**
- **Clinical Rationale: Day 0 claims represent discharge meds**
(e.g., antiemetics, antibiotics, analgesics) prescribed AFTER the visit
- **Prevents immediate reverse-causality target leakage**

Phase 4: 1:5 Matched Control Sampling (phase3_cohort_creation.py)

Eligible Control Pool
- **Covered APCD members in same partition year**
- **Zero qualifying ED visits in study year**
- **Verified health system contact continuity**

1:5 Incidence-Density Matching (No Replacement)
Exact Stratification On:
1. **Age Band (8 Strata: 0–12, 13–24, 25–44, ..., 85–114)**
2. **Biological Sex (Exact M/F Match)**
3. **Calendar Year Partition (2016–2019)**
4. **Active Coverage Continuity**

Pseudo-Index Reference Date (t_pseudo)
Controls assigned earliest non-ED medical claim date
Guarantees identical 21-day pre-index baseline window

Phase 3: Acute Exposure Horizon & Feature Engineering

21–30 Day Acute Pre-Index Window
[t_0 - 21/30 days → t_0 - 1 day]
- **Active Multi-Drug Concurrency & Drug-Drug Interactions**
- **Beers Criteria High-Risk Medications in Older Adults**
- **Anticholinergic & Sedative Cumulative Burden**
- **Sudden Titrations & Recent Regimen Modifications**

Toolbox Demarcation Architecture
- **Tabular Counts & Claims-Density Bins (P25/P50/P95) → Tree Feature Matrix**
- **bupaR Process Flows, DTW Clusters & FP-Growth Rules → Dashboard Only**
- **Locked 682-Feature Pre-Index Whitelist Enforced**

Phase 5: QA Assertion & Checkpointed Export (phase4_finalization.py)

```
DuckDB QA Unit Assertion (qa_index_date_uniqueness.py)
SELECT mi_person_key, count(DISTINCT first_ed_non_opioid_date)
HAVING count > 1;
Asserts EXACTLY ONE index date per person-year
```

Cell-Size Suppression Guard
Suppresses n ≤ 10 cell counts per CMS/VHI privacy policy

Final Parquet Fact Table (s3://pgx-repository/cohorts/)
- **schema: final_cohort_schema.json**
- **Cases + 5:1 Controls with 682 Whitelisted Features**
- **Checkpointed to S3 pgx-pipeline-status/**